

UTKARSH BARNWAL

Greater Noida, Uttar Pradesh

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EDUCATION

Lloyd Institute of Engineering and Technology

Bachelor of Technology (B.Tech) CGPA: 7.8

Jun 2023 – Jun 2026

Greater Noida, Uttar Pradesh

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript, Python

Frontend: React.js, Next.js, Tailwind CSS, Redux, Zustand

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Databases: MongoDB, Redis, SQL, Pinecone (Vector DB)

DevOps & Tools: Docker, Nginx, Git, GitHub, AWS (Basics)

AI & Systems: RAG, LLM APIs (OpenAI, Gemini), Custom AI Agents, Microservices, RabbitMQ

PROJECTS

Aurora – LLM Powered RAG System | MERN, RAG, Gemini API, Vector DB

June 2025

- Built a full-stack **LLM-powered assistant** using Retrieval-Augmented Generation (RAG) for contextual responses over custom datasets.
- Engineered **vector embeddings and semantic search** using Pinecone to improve retrieval accuracy by reducing hallucinations.
- Designed scalable backend with **Node.js & Express**, optimizing response latency for AI queries.
- Integrated **JWT authentication** and session management for personalized AI interactions.

E-Commerce Backend with AI Agent | Node.js, Express, MongoDB, Redis, Docker, REST APIs July 2025

- Architected a **microservices-based backend** handling products, cart, orders, authentication and payments.
- Developed 20+ RESTful APIs with structured MongoDB schemas for efficient data querying.
- Optimized performance using **Redis caching** reducing repeated database calls and improving response performance.
- Built an **AI automation agent** simulating user behavior (cart actions, session workflows).
- Containerized services using Docker and enforced **RBAC with JWT authentication**.

HACKATHONS

Reimagine Hackathon

2025

Frontend Hackathon by Sheryians

- Developed responsive UI clones using React and Tailwind CSS with reusable component architecture. with a focus on **UI/UX**, responsiveness, and modern frontend practices.
- Implemented scalable and reusable components using **ReactJS, TailwindCSS, ShadCN UI**.
- Enhanced accessibility and performance, improving user experience significantly.

Phantom Agents Hackathon

2025

AI Hackathon

- Designed modular AI agents for autonomous workflow execution using LangChain and LLM APIs. and enhancing decision-making.
- Integrated **LangChain, Vector Databases (Pinecone/Redis)** and **LLMs** for context-aware responses.
- Designed modular agent workflows for scalability and multi-agent communication.

CERTIFICATIONS

- Frontend Development (React, Tailwind CSS) – Sheryians
- Backend Development (Node.js, Express.js, MongoDB) – Sheryians
- Projects & Hackathons – Sheryians